D 156 | Ownership needs a denominator

FICTIONAL offline economics lab. No actual issuer, investor, financing, term sheet, legal clearance, account or spend.

Three independent rounds	Crore: 1.5 new /13.5 pre, 8/32, 40/160. Pre includes all prior as-converted holdings. No options, convertibles, fees, secondary or pro rata. Founder .576, round 1 .064, seed .16, A .20. Stipulated prices are not market evidence.
Pool ordering	Post-A pool target .10 dilutes everyone: .5184/.0576/.144/.18/.10. Separate protected-A scenario carves .10 final pool out of pre-A holders while A stays .20: .504/.056/.14/.20/.10. No existing pool; fractional ownership, not issued grants or voting shares.
Conditional cap stack	Two cap-binding checks 1/20 and 1/25 imply .05+.04 before priced money. New money fraction .20 gives SAFE 1 .04, SAFE 2 .032, founder .728, new .20. This is not a full recursive actual-form solver. Sum below 1 is necessary, not proof the cap binds.
Price alternatives	Separate supplied capitalization toy: purchase 1 crore, cap 20 crore, capitalization 1000000 shares =>cap price 200. Supplied standard price 250 gives 50000 cap shares versus 40000 standard, choose 50000. At 150 standard gives 66666.666667, greater than 50000. Denominator must already reconcile all converting securities.
Simple note	Principal 1 crore, annual.08, 18 months, 20% discount on 100/share =>interest.12, total 1.12 crore, price 80, 140000 shares. Toggle whether interest converts. Excludes compounding, rounding clauses, maturity remedies and real contract suitability.
Scope before action	Actual Indian issuer eligibility, residence, beneficial ownership, sector, deposit, FEMA, tax, approvals/reporting and operative document version unresolved/unscored. US template labels and iSAFE names prove none of those. No real person/contact/signature/wire/account.

Round inputs | No overhang, no existing pool

Shaded cells are fictional inputs. Independent families; invalid family must not silently contaminate unrelated arithmetic.

Round 1 new money crore	1.50
Round 1 pre-money crore	13.50
Seed new money crore	8.00
Seed pre-money crore	32.00
A new money crore	40.00
A pre-money crore	160.00
Final target pool fraction	0.10

Numeric/domain gate (1=yes)	1.000000

Pre-money includes all prior as-converted holdings. Zero new money allowed; positive pre required. Target pool must also fit below existing-holder fraction in protected-A branch. This is dilution, not liquidation preference, enterprise valuation or executed ownership.

Round walk | Fractions of fully diluted total

No pool in this walk. Each round dilutes all prior holders. Totals must reconcile to 1.

Holder	After round1	After seed	After A	Post-A pool
Founders	0.900000	0.720000	0.576000	0.518400
Round1	0.100000	0.080000	0.064000	0.057600
Seed	0.000000	0.200000	0.160000	0.144000
A	0.000000	0.000000	0.200000	0.180000
Pool	0.000000	0.000000	0.000000	0.100000
Total	1.000000	1.000000	1.000000	1.000000

Baseline fractions: afterA .576/.064/.16/.20/0; post-A pool .5184/.0576/.144/.18/.10. Adding dilution percentages is wrong. A target 10% final pool requires new pool shares equal to old total*.10/.90, not old total*.10. Reserved pool is a fully diluted convention, not issued ordinary stock.

Protected pool | A investor stays protected

No existing pool. Same final target, different burden. Pre-A holders own 1 - A fraction - pool fraction after close.

A fraction without post-close pool	0.200000
Target final pool	0.100000
Feasible positive pre-A holder remainder	1.000000
Retention factor for pre-A holders	0.875000
Founders final	0.504000
Round1 final	0.056000
Seed final	0.140000
A final	0.200000
Pool final	0.100000
Total	1.000000
Founders protected minus post-A pool	-0.014400

Baseline founders.504, round 1.056, seed.14, A.20, pool.10, total 1; founder difference-.0144 versus pro-rata post-A issuance. Target must be below 1-A to leave positive old ownership. This defined branch is not every term sheet or an incremental top-up with an existing pool.

Cap inputs | Conditional cap-binding branch

Shaded cells are fictional inputs. Independent families; invalid family must not silently contaminate unrelated arithmetic.

SAFE1 purchase crore	1.00
SAFE1 post-money cap crore	20.00
SAFE2 purchase crore	1.00
SAFE2 post-money cap crore	25.00
Priced new money final fraction	0.20

Numeric/domain gate (1=yes)	1.000000

No other securities/options/pool. This branch assumes both caps actually bind. It does not decide that from only two caps and checks. Unknown five-check allocations stay unknown. At cap sum 1 or above, branch undefined rather than invented negative founders.

Cap walk | Ratios are not locked forever

The gate is arithmetic feasibility, not an operative-form conversion verdict. No interest/maturity imported into SAFE.

SAFE1 before new money	0.050000
SAFE2 before new money	0.040000
Combined SAFE before	0.090000
Founders before	0.910000
SAFE1 after	0.040000
SAFE2 after	0.032000
Founders after	0.728000
Priced investors after	0.200000
Total after	1.000000

Baseline .05/.04/.09/.91 before; .04/.032/.728/.20 after. Sum 100%. Pro rata would require new cash and separate agreed right; no free percentage protection. Five-check sum needs each allocation/cap and conversion branch, not endpoints or average cap.

Price inputs | Supplied reconciliation, not solver

Shaded cells are fictional inputs. Independent families; invalid family must not silently contaminate unrelated arithmetic.

Purchase crore	1.00
Cap crore	20.00
Supplied reconciled capitalization shares	1000000.00
Supplied standard price rupees/share	250.00
Alternative supplied standard price rupees/share	150.00

Numeric/domain gate (1=yes)	1.000000

Separate illustrative share comparisons. Capitalization includes this and other converting securities as defined by form; do not treat 1000000 as founder-only shares. Standard price is supplied, not derived from an independent valuation. No circular cap-table solve or ownership inferred here.

Price and note | Independent families

Crore is 10000000 rupees. Share counts may be fractional in this math toy; actual rounding terms not assumed.

Cap price rupees/share	200.000000
Cap-price share count	50000.000000
Standard-price share count	40000.000000
Greater share count	50000.000000
Alternative standard-price share count	66666.666667
Alternative greater count	66666.666667

Baseline cap price 200/shares 50000; standard 250=>40000, choose 50000. Alternative 150=>66666.666667, choose alternative. Supplied capitalization must be re-reconciled if terms change; fixed-capitalization sensitivity is not a fully solved low-round company.

Note principal crore	1.00
Annual simple interest fraction	0.08
Elapsed months	18.00
New-money price rupees/share	100.00
Discount fraction	0.20
Interest converts flag 0/1	1.00

Note numeric/domain gate	1.000000

Accrued simple interest crore	0.120000
Principal plus accrued interest crore	1.120000
Discounted conversion price rupees/share	80.000000
Amount converting crore	1.120000
New shares	140000.000000

No term sheet. Interest flag 0 excludes accrued interest from shares, not erasing a real repayment obligation. Baseline flag 1: .12/1.12/80/1.12/140000. Flag 0=>125000 shares. Zero principal/months/rate allowed. No compound interest or maturity remedy inferred.

Instrument evidence | Comparison, not contract

Actual templates used for interpretation only. No actual issuer, governing law, tax opinion or completed instrument.

Cap-only Sections 1-2	Greater share count from purchase/standard price or purchase/Safe Price. Company Capitalization includes converting securities, promised options and defined pool, with increase exception. Liquidity Capitalization differs. No ordinary voting stock before issuance; defined dividend/event rights remain.
MFN Section 3	Retrieved actual form: company notice and copy; investor written election within 10 days after receipt; then matching amendment. Defined exclusions include certain side letters/options/commercial securities. Not automatic cherry-picking. Current landing shorthand conflicts; operative version matters.
No maturity	SAFE not a standard interest-bearing note. No maturity does not erase event proceeds. Cash-out priority behind creditors and alongside specified securities; insufficient legally available proceeds can mean less than purchase. Entitlement is not collection guarantee.
Priced and notes	Priced round ownership follows fully defined capitalization/terms. Notes need interest convention, conversion of interest, price/discount/cap, events/maturity and remedies. Neither delay-proportional discount nor control concession assumed. Pro rata is optional agreement and new money.
Indian applicability	Unknown issuer jurisdiction, investor residence/beneficial ownership, startup recognition, sector, instrument eligibility, deposit, FEMA, valuation, approvals, reporting, tax, current amendments and legal review. Selected prior NDI/playbook text is not a complete current law review. Leave legality unscored.
Transfer checks	Change A new money 40 to 80: A fraction 1/3, founders.48 beforepool; postpool.432; protectedpool.408. Change note months 24=>.16 interest/145000 shares. Change cap 1 from 20 to 10=>combined.14 before/.112 after. Static prose baseline only; refresh before changed export.

Actual evidence | Blank is unknown

No real cap table, owner, signature, account, grant, investor contact, wire, financing document or legal clearance.

Case / terms	Source / version	Capitalization basis	Legal / tax question	Permission / audience	Accepted reviewer	Next check / evidence

Actual instrument and Indian applicability remain unscored. Authored tests are not learner transfer, independent legal review, native Excel/Sheets parity, full source AV or overall 10. Blank reviewer is not consent or accepted responsibility. No calendar or monitor created.

Evidence | Primary forms before shorthand

Seven URLs/three publishers/three source types: official forms/guide, practitioner explanation, product documentation. No independent empirical validation.

YC cap-only actual form	Complete returned paragraph text read. Sections 1-2 event rights/capitalization and US representations/tax language. Table/text-box content and fresh source-page pixels not reviewed here; earlier full rendered-form evidence reused. No completed form or legal permission.
YC MFN actual form	Complete returned paragraph text read. Section 3 explicit 10 day written election and defined exclusions beat automatic landing copy. US-specific representations/tax intention are not Indian rules. Version/execution controls actual transaction.
YC guide and landing	Current returned guide QuickStart/intro/Q&A A-F selected sections throughF 6 read; not fresh full 33 pageappendices/pixels. Low/near-cap exception and optional pro rata. Current landing written read; agent onboarding excluded from all execution. Guide contains generalizations/version residue.
Carta option pool	Complete returned written Aug 20, 2026 article read, not images/linked benchmarks. Pre-money versus post-money burden supports defined scenarios. US 409 A/Rule 701 not imported into India. Generic 10-15% and reported 71%/616 days not validated or adopted as this toy hiring requirement.
Carta release definitions	Complete returned older product note read, not current product execution/price/access. Formula labels and formatted dilution definition imperfect; independent arithmetic controls. No claim actual calculator or service was run.
Founder Decisions	Complete returned written explanation read, not interactive UI or linked Kruze source. Same-target pre/post distinction; its generic convertible-pool shorthand is not universal. ActualYCdefined-capitalization mechanics take precedence. No actual local-data/privacy behavior independently tested.

Source 1 | open verified source

Source 2 | open verified source

Source 3 | open verified source

Source 4 | open verified source

Source 5 | open verified source

Source 6 | open verified source

Source 7 | open verified source

Failure log | Controlled authored tests

Independent arithmetic/changed-input/error cases only; no actual financing, learner, account or law validation.

1	PASS: 59 baseline formulas independently recomputed.
2	PASS: 23 changed sets/1357 checks.
3	PASS: 12 boundaries/708 checks.
4	PASS: 20 invalid sets/1180 checks.
5	PASS: 6 broken references propagate errors.
6	PASS: independent round/cap/price/note families isolated.
7	PASS: undefined infeasible pool/cap/zero-price cases.
8	PASS: blank actual evidence;not legal/native parity/learner.